Data Science with Python
Ticket Out the Door
Set 17: Python Strings

Name ____ Period ______

Skill 17.1 Exercise 1

Refer to the string below,

```
pet = "Sparky the dog"
```

Position	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Letter	S	p	a	r	k	y		t	h	e		d	o	g

For each statement, indicate what is printed.

`letters = pet[3]` `print(letters)`	
`letters = pet[len(pet)]` `print(letters)`	
`letters = pet[len(pet)/2]` `print(letters)`	
`letters = pet[len(pet)/3]` `print(letters)`	

Skill 17.2 Exercise 1

You need to write a program that generates usernames for students at Timberline. The usernames will use the following convention: first_initial + last_name. For example, Lady Gaga would have the following username, LGaga.

Write a function called *make_un* that could be used to create a username. Your function should accept two parameters (one for the first name and one for the last name) and return the username.

In addition to usernames, the students also need passwords. The default passwords must use the following convention: first letter first name + first letter lastname + 6 digit birthdate (EX: LG032886)

Write a function called *make_pw* that could be used to create a password. Your function should accept three parameters (one for the first name, one for the last name, and one for the birthdate (MM/DD/YYYY)) and return the password.

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Consider the following student information, first_name = “Lady” last_name = “Gaga” birthday = “03/28/1986” Call each of the functions above using the student information.

Skill 17.2 Exercise 2
Temporary passwords are needed in the event a user forgets their password. Write a function called *temp_pw* that has two parameters, one for the first name and one for the last name. Your function should combine the last three letters of the first name with the last three letters of the last name and return the result.

Skill 17.3 Exercise 1

What is wrong with the code below,
 	
`word = "flueberry"`
 	
`word[0] = "b"`

Fix the code so that the first letter in *word* is changed to “b”.

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Write a function called *scramble_word* that accepts a parameter. In the body of *scramble_word* scramble the word by swapping the first and last letter of the word. For example, the following call word print *ylueberrb* `print(scramble_word(“blueberry”))`

Skill 17.4 Exercise 1
Write code that could be used to print the alphabet to the console exactly as shown – using just ONE console.log statement. A   B C   D
Write code that will print the following to the console. “Woo Hoo!” Note, the quotes must display when printed.

Skill 17.5 Exercise 1
Write a new function called *get_length()* that takes a string as an input and returns the number of characters in that string. Do this by iterating through the string. Do not use the len() function!

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Skill 17.5 Exercise 2
Write a function called *letter_check* that takes two inputs, word and letter. The function should return True if the word contains the letter and False if it does not.

Skill 17.6 Exercise 1
Write a function called contains that takes two arguments, big and little. The function should return true if big contains little. For example contains("watermelon", "melon") should return True and contains("watermelon", "berry") should return False.

Skill 17.6 Exercise 2
Write a function called *common_letters* that takes two arguments, *string_one* and *string_two* and then returns a list with all of the letters they have in common. The letters in the returned list should be unique. For example, `print(common_letters('manhattan', 'san francisco'))` should return [a, n]